

# DTSC: A Dual-Tier Sector-Based Cooperative Mobile Charging Scheme for Wireless Rechargeable Sensor Networks

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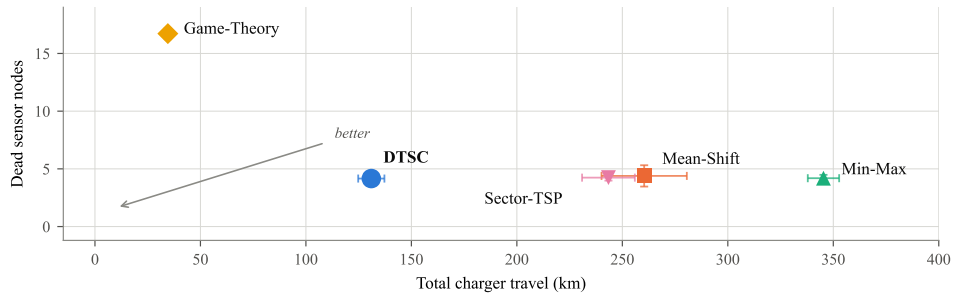
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**Problem.** In wireless rechargeable sensor networks, energy demand is uneven: nodes near the base station relay traffic for everyone else and drain fastest, while distant nodes are the most expensive to reach. Chargers scheduled independently over the whole field therefore burn much of their budget on repeated long-distance return trips.

**Approach.** DTSC estimates each node's energy consumption rate and partitions the field into sectors of balanced workload. Each sector is served by an *inner–outer charger pair*: the outer charger never returns to the base station; the inner charger refuels there and meets it at a rendezvous point to transfer energy by contact. The long return leg is thus paid once per pair, not once per charger.

**Results.** Across 12,500 simulations (125 settings  $\times$  100 seeds), DTSC at the heaviest load **travels 46–62% less than Sector-TSP, Mean-Shift and Min-Max while losing no more nodes**: 4.16 dead against 4.19, 4.24 and 4.39, all within one standard error. Game-Theory travels least, serving only urgent requests, at the cost of **4 $\times$  more dead nodes** (16.7 vs. 4.2)—a gap over twenty standard errors wide. DTSC's travel also scales sub-linearly, growing 2.2 $\times$  from 500 to 1500 nodes where Mean-Shift's grows 13 $\times$ .



**Fig. 1.** Travel–reliability trade-off at 1500 nodes; bars are  $\pm 1$  standard error over 100 seeds. DTSC is level with three baselines on dead nodes at roughly half their travel; only Game-Theory travels less, and it loses four times as many nodes.

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- [2] A. Kaswan *et al.*, “A survey on mobile charging techniques in WRSNs,” *IEEE Commun. Surveys Tuts.*, 24(3), 1750–1779, 2022.